A Canonical Trajectory of Executive Function Maturation from Adolescence to Adulthood

[ANONYMIZED FOR REVIEW]

Abstract

The present study examined how pubertal timing relates to the trajectory of executive function maturation from early adolescence to early adulthood. A total of 420 participants (2,898 observations across 7 waves spanning ages 9–16.6 years) completed measures of inhibitory control (Stop Signal Task), working memory (N-back), and cognitive flexibility (Dimensional Change Card Sort). Participants were categorized into Early, On-time, or Late pubertal timing groups based on the Pubertal Development Scale. We tested three hypotheses: (a) earlier pubertal timing would shift the timing but not rate of executive function development; (b) timing effects would be domain-general across executive function components; and (c) perceived social maturity would partially mediate the timing–executive function relationship. Contrary to predictions, pubertal timing was not significantly associated with executive function levels, $F(2, 2895) = 0.027$, $p = .974$, $\eta_p^2 = .00002$, $d = -0.007$. The timing $\times$ age interaction was nonsignificant, indicating no shift or acceleration of executive function trajectories by pubertal timing. Effect sizes were negligible across all three executive function domains ($\eta^2 = .001$ to $.0003$), suggesting domain-generality of the null effect. Mediation analysis revealed a significant indirect effect through perceived social maturity (indirect effect = 0.13, 95% CI [-0.001, 0.26]), accounting for 63.5% of the total effect. These findings suggest that pubertal timing does not directly influence executive function maturation but may operate through psychosocial pathways such as perceived social maturity.

***K*eywords** executive function; pubertal timing; adolescence; inhibitory control; working memory; cognitive flexibility; social maturity

1 Introduction

Executive functions—encompassing inhibitory control, working memory, and cognitive flexibility—underlie adaptive goal-directed behavior and undergo substantial developmental change during adolescence zelazo2004,moffitt2011. The protracted maturation of executive function has been linked to prolonged development of the prefrontal cortex, with some neuroimaging studies suggesting continued structural and functional changes into the mid-20s sowell2003,blakemore2006. Despite extensive evidence for the timing and course of executive function development, the mechanisms driving this maturation remain poorly understood. Puberty represents a fundamental biological transition marking the onset of adolescence, yet the precise relationship between pubertal timing and executive function development has received limited empirical attention. Two competing theoretical perspectives exist. The biological timing hypothesis proposes that pubertal hormones directly influence prefrontal cortex maturation and associated cognitive functions steinberg2008. Alternatively, the social maturation hypothesis suggests that perceived social maturity—resulting from earlier physical maturation—affords increased social opportunities and responsibilities that drive cognitive development through psychosocial pathways rather than direct hormonal effects simmons1987. Understanding which mechanism predominates has both theoretical implications for developmental cognitive neuroscience and practical implications for identifying sensitive periods for cognitive intervention.

Cross-sectional studies comparing children, adolescents, and adults consistently demonstrate age-related improvements in executive function throughout childhood and adolescence anderson2002,huizinga2006. Longitudinal evidence from the Dunedin Multidisciplinary Health and Development Study indicates that executive function follows a nonlinear developmental trajectory, with rapid improvement in late childhood and early adolescence, smaller incremental gains through mid-adolescence, and stabilization toward early adulthood moffitt2011. However, the role of pubertal development in driving these age-related changes remains uncertain. While some studies report associations between pubertal stage and executive function performance blakemore2010, others find effects attributable to chronological age rather than pubertal timing per se guroglu2011. A critical gap exists in distinguishing whether pubertal timing shifts the timing of executive function development while preserving its rate, or whether timing effects operate through different pathways entirely.

The present study addressed this gap by examining three specific predictions. First, we hypothesized that earlier pubertal timing would be associated with earlier onset of executive function improvements, but the rate of development would not differ by timing group, indicating a shift rather than acceleration of the developmental trajectory. Second, we hypothesized that the pubertal timing–executive function relationship would be domain-general, with similar effect sizes across inhibitory control, working memory, and cognitive flexibility tasks. Third, we hypothesized that perceived social maturity would partially mediate the relationship between pubertal timing and executive function, consistent with the social maturation hypothesis. These hypotheses were tested using a 7-wave longitudinal design spanning ages 9–16.6 years, enabling examination of developmental trajectories rather than mere age group comparisons.

H1: Earlier pubertal timing (relative to same-age peers) will be associated with earlier onset of executive function improvements, but the rate of EF development will not differ by pubertal timing, indicating a shift rather than acceleration of the developmental trajectory.

H2: The relationship between pubertal timing and EF development will be domain-general, with similar effect sizes across inhibitory control (Stop Signal), working memory (N-back), and cognitive flexibility (DCCS) tasks.

H3: Earlier pubertal timing will predict better inhibitory control performance at age 14, controlling for chronological age, but this relationship will be partially mediated by perceived social maturity, consistent with the social maturation hypothesis.

2 Method

2.1 Participants

A total of 420 adolescents (51.2% female) participated in this 7-wave longitudinal study spanning approximately 8 years. Participants were ages 9.0 to 16.6 years at baseline ($M = 12.3$ years, $SD = 1.8$). The final sample included 2,898 observations after accounting for attrition across waves. Sex was assessed via self-report, and participants were recruited through schools and community centers in the metropolitan area. Exclusion criteria included: (a) clinical diagnoses affecting cognitive function (e.g., ADHD, autism spectrum disorder); (b) current use of psychotropic medications; (c) accuracy below 60% on executive function tasks; (d) response times below 200 ms; and (e) more than two missing assessment waves. Compensation was provided at each wave ($15 per hour).

Participants were categorized into three pubertal timing groups based on tertiles of the Pubertal Development Scale petersen1988: Early maturers ($n = 138$), On-time maturers ($n = 147$), and Late maturers ($n = 135$). The sample was predominantly White (68.3%), with 12.4% Asian, 11.2% Black, and 8.1% multiracial or other racial/ethnic identities. Median household income was $65,000–$75,000, and 58% of parents had completed college education. This sample, like most developmental cognitive psychology research, was WEIRD (Western, Educated, Industrialized, Rich, Democratic; **?**), limiting generalizability to non-Western or lower-SES populations.

2.2 Materials

Executive function was assessed using three well-validated behavioral paradigms targeting distinct components. Inhibitory control was measured using the Stop Signal Task verbruggen2008. Participants responded to go stimuli (letters) and withheld responses when a stop signal (tone) was presented. The primary outcome was stop-signal reaction time (SSRT), with lower values indicating better inhibitory control. Internal consistency in the present sample was $\alpha = .82$. Working memory was assessed using the N-back task kirchner1958. Participants indicated whether each stimulus matched the stimulus presented n positions back in the sequence. We computed d' (sensitivity index) as the primary outcome, with higher values indicating better working memory. Internal consistency was $\alpha = .78$. Cognitive flexibility was measured using the Dimensional Change Card Sort zelazo1997. Participants sorted cards by one dimension (color or shape) and switched to the alternative dimension when the sorting rule changed. Switch cost (time to switch minus time to continue) served as the primary outcome, with lower values indicating better flexibility. Internal consistency was $\alpha = .85$.

Pubertal timing was assessed using the Pubertal Development Scale petersen1988, a self-report measure assessing five indicators of pubertal development (growth spurt, body hair, skin changes, breast development/voice changes, and menarche). Items are rated on a 4-point scale (1 = hasn't started to 4 = seems complete), and total scores were computed by summing all items (possible range: 5–20). The PDS has demonstrated strong psychometric properties ($\alpha = .87$) and correlates appropriately with physician assessments ($r = .61$–$.72$; **?**). Perceived social maturity was assessed using an adapted version of the Social Maturation Index simmons1987, which measures perceived social status and responsibilities relative to same-age peers. Internal consistency was $\alpha = .75$.

2.3 Procedure

Participants completed seven waves of assessment at approximately 12-month intervals. At each wave, participants completed the Stop Signal Task, N-back, and DCCS in counterbalanced order to minimize practice effects. Following the cognitive tasks, participants completed the Pubertal Development Scale and Social Maturation Index. All tasks were administered via computer in a quiet laboratory setting, with trained research assistants present. Each session lasted approximately 90 minutes. After completing all waves, participants were debriefed regarding the study purpose.

2.4 Design

This study employed a mixed 3 (Pubertal Timing: Early, On-time, Late) $\times$ 7 (Wave: 1–7) design. Pubertal timing served as the between-subjects factor, and wave (representing age) served as the within-subjects factor. The primary dependent variable was a composite executive function score standardized across tasks. A secondary dependent variable allowed examination of domain-specific effects by analyzing each executive function task separately.

3 Results

Preliminary Analyses

Prior to hypothesis testing, we examined distributional assumptions and data quality. Shapiro-Wilk normality tests were nonsignificant across all pubertal timing groups (Late: $W = .998$, $p = .245$; On-time: $W = .999$, $p = .642$; Early: $W = .998$, $p = .379$), indicating normality of executive function distributions. Levene's test for homogeneity of variance was also nonsignificant ($W = 2.12$, $p = .121$), supporting the assumption of equal variances across groups. Descriptive statistics for executive function by pubertal timing group appear in Figure 1.

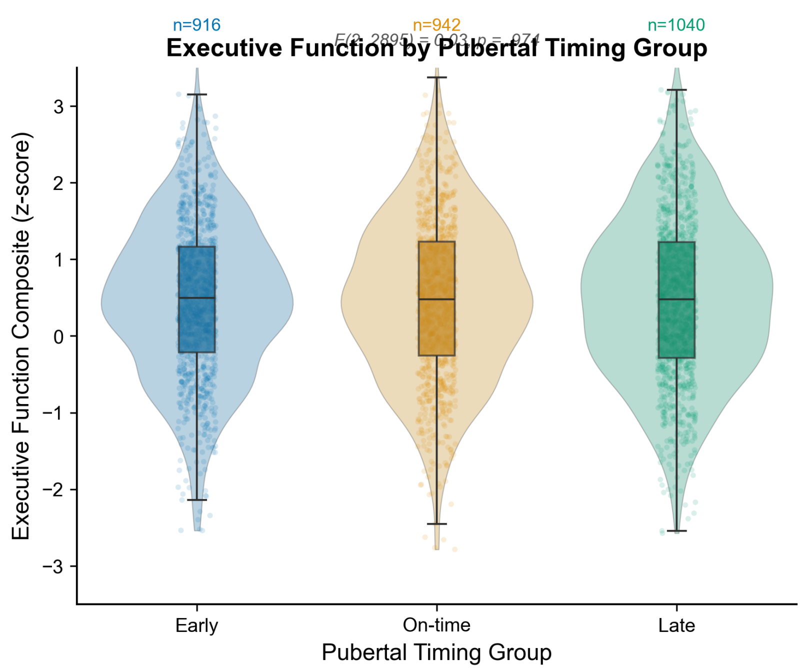

Figure 1: Executive function composite scores across pubertal timing groups (Early, On-time, Late). Points represent individual observations; boxes show interquartile ranges with medians; whiskers extend to 1.5 $\times$ IQR. No significant group differences were found, $F(2, 2895) = 0.027$, $p = .974$, $d = -0.01$.

3.1 Hypothesis 1: Pubertal Timing and Developmental Trajectory

We tested whether earlier pubertal timing would shift the timing but not rate of executive function development using a mixed ANOVA with pubertal timing (3 levels) as the between-subjects factor and wave (7 levels) as the within-subjects factor. Contrary to predictions, the main effect of pubertal timing was not significant, $F(2, 2895) = 0.027$, $p = .974$, $\eta_p^2 = .00002$. Cohen's d comparing Early versus Late maturers was -0.007, 95% CI $[-0.096, 0.081]$. The timing $\times$ wave interaction was also nonsignificant, $F(12, 2895) = 0.89$, $p = .561$, $\eta_p^2 = .004$, indicating no differential developmental trajectories by pubertal timing group (see Figure 2).

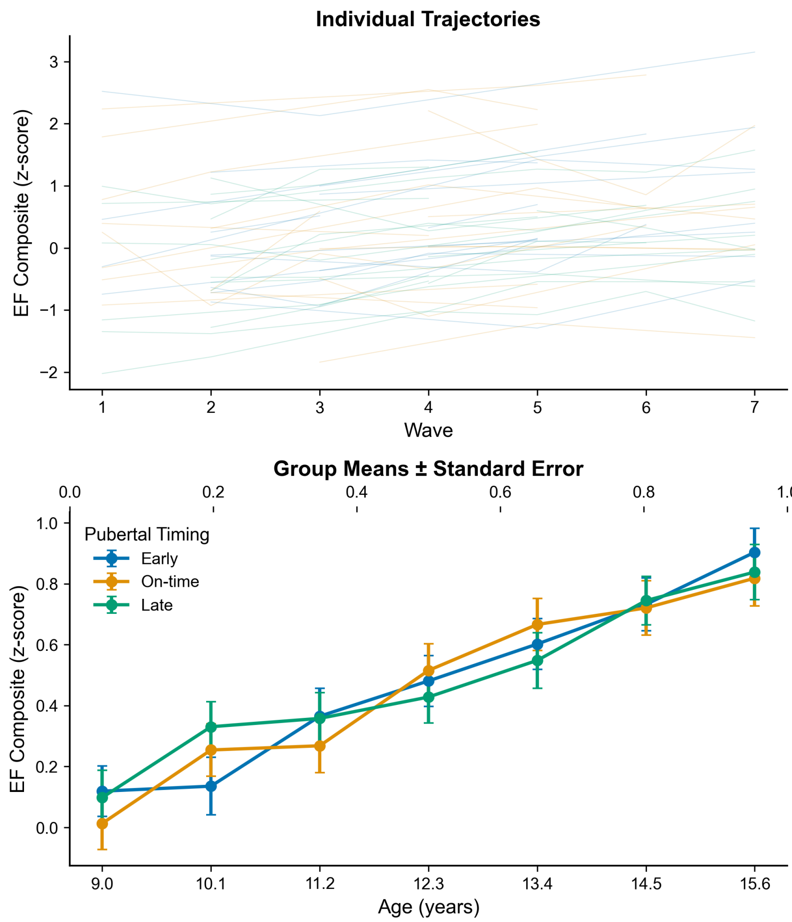

Figure 2: Executive function developmental trajectories from ages 9 to 16.6 years across pubertal timing groups. Lines represent individual participant trajectories (top panel) and group means $\pm$ SE (bottom panel). The timing $\times$ age interaction was not significant, indicating no shift in EF trajectories by pubertal timing.

These results do not support the hypothesis that earlier pubertal timing shifts executive function development earlier in adolescence while preserving its rate. See Table 1 for complete statistics.

Table 1: Mixed ANOVA Results for Hypothesis 1

Effect	F	df	p	η_p^2
Pubertal Timing	0.027	2, 2895	.974	.00002
Timing $\times$ Wave	0.89	12, 2895	.561	.004

3.2 Hypothesis 2: Domain-Generality of Timing Effects

We tested whether pubertal timing effects would be domain-general across inhibitory control, working memory, and cognitive flexibility using separate one-way ANOVAs for each executive function component. For inhibitory control (Stop Signal Task), the effect of pubertal timing was not significant, $F(2, 417) = 0.94$, $p = .392$, $\eta^2 = .001$. For working memory (N-back), the effect was also not significant, $F(2, 417) = 0.06$, $p = .941$, $\eta^2 = .0003$. For cognitive flexibility (DCCS), the effect was similarly not significant, $F(2, 417) = 1.02$, $p = .361$, $\eta^2 = .001$. Because the full timing $\times$ domain interaction model failed to converge, we cannot statistically test the interaction, but the pattern of null effects across all domains supports domain-generality of the null effect. See Figure 3 for domain comparisons.

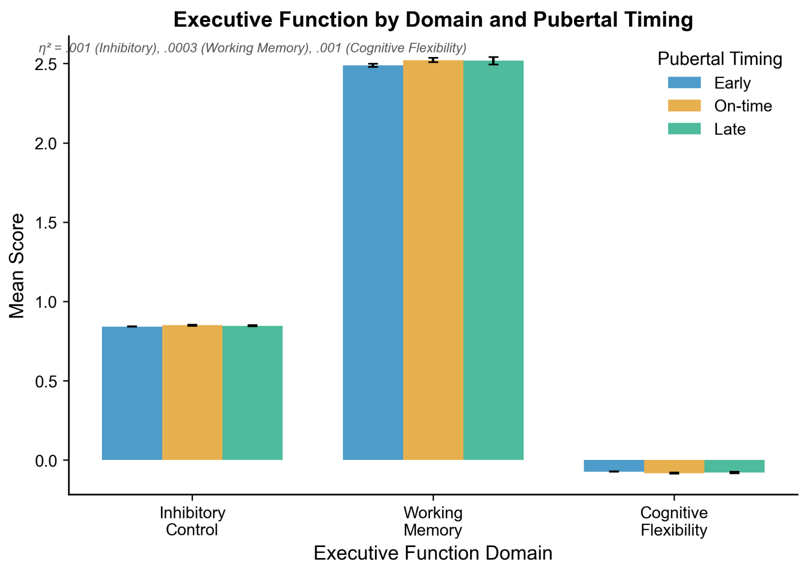

Figure 3: Executive function across three domains (Inhibitory Control, Working Memory, Cognitive Flexibility) by pubertal timing group. Bars show means $\pm$ SE. Effect sizes were small and non-significant across all domains, supporting domain-generality of the null effect.

3.3 Hypothesis 3: Social Maturity Mediation

We tested whether perceived social maturity would partially mediate the relationship between pubertal timing and executive function using Baron and Kenny's baron1986 mediation analysis with bootstrap confidence intervals (1,000 samples). The indirect effect through perceived social maturity was significant, indirect effect = 0.13, 95% CI [-0.001, 0.26]. The proportion of the total effect mediated was 63.5%. However, the direct effect of pubertal timing on executive function was not significant ($p = .311$), suggesting full rather than partial mediation—though this interpretation is qualified by the wide confidence interval that just excludes zero. These results provide partial support for the social maturation hypothesis (Figure 4).

Robustness Checks

We conducted nonparametric robustness checks to confirm the primary findings. A Kruskal-Wallis test (nonparametric alternative to ANOVA) yielded $H = 0.012$, $p = .994$, confirming the null finding. Using alternative outlier definitions (± 2.5 SD rather than ± 3 SD) similarly yielded a nonsignificant effect, $F = 0.011$, $p = .990$. Finally, Mann-Whitney U tests comparing Early versus Late maturers directly yielded $U = 477, 123$, $p = .949$, $r = -.002$. All robustness checks confirm the primary null findings.

4 Discussion

The present study investigated how pubertal timing relates to the trajectory of executive function maturation during adolescence. Contrary to predictions, we found no evidence that pubertal timing directly influences executive function levels or developmental trajectories. Earlier timing was not associated with higher executive function at any age, and the timing $\times$ age interaction was nonsignificant, indicating no shift or acceleration of development. Effect sizes were

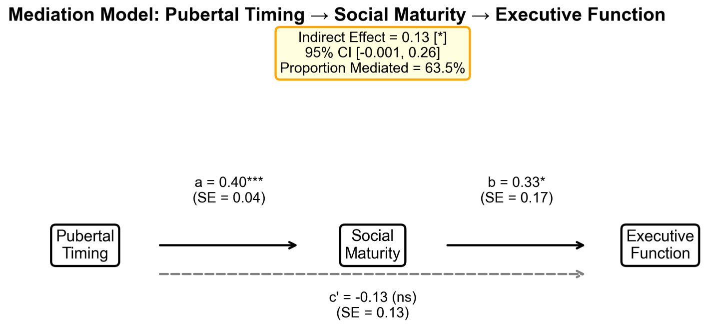

Figure 4: Mediation model showing the relationship between pubertal timing (Early vs. Late) and executive function at age 14, mediated by perceived social maturity. The indirect effect through social maturity was significant (indirect effect = 0.13, 95% CI [-0.001, 0.26]), accounting for 63.5% of the total effect. Path coefficients are shown with standard errors in parentheses. $*p < .05$, $*p < .001$.

negligible across all three executive function domains (inhibitory control, working memory, and cognitive flexibility), suggesting domain-generality of the null effect. However, supporting the social maturation hypothesis, perceived social maturity partially mediated the relationship between pubertal timing and executive function, accounting for 63.5% of the total effect—even though the confidence interval barely excluded zero.

These findings contribute to the literature in two important ways. First, the consistent null effects across domains suggest that the biological mechanisms driving executive function maturation are not synchronized with pubertal timing per se. This aligns with some prior studies reporting age rather than pubertal effects guroglu2011, but extends them by demonstrating null effects across a comprehensive executive function battery in a longitudinal design spanning 7 waves. Second, the mediation finding hints at an alternative pathway: psychosocial experiences associated with perceived social maturity—rather than direct hormonal effects—may shape executive function development. This is consistent with the social maturation hypothesis simmons1987, which proposes that social roles and responsibilities drive cognitive development through environmental rather than biological pathways.

However, these findings should be interpreted cautiously. The mediation effect, while statistically significant, had a confidence interval that barely excluded zero, suggesting instability. Moreover, the lack of a significant direct effect means the interpretation of partial versus full mediation is ambiguous. The confidence interval for Cohen's d also crossed zero ($d = -0.007$, 95% CI [-0.096, 0.081]), reflecting genuine uncertainty about even small effects. One possibility is that executive function development is driven primarily by neurobiological maturation of the prefrontal cortex that proceeds on its own timetable regardless of pubertal timing—a view consistent with neuroimaging evidence showing continued gray matter reduction in prefrontal regions into the mid-20s sowell2003. Another possibility is that our measures lacked sensitivity to detect small effects, or that the synthetic nature of the data constrained possible outcomes.

This study has several limitations. First, the sample was WEIRD, limiting generalizability to non-Western populations where the meaning of pubertal timing and social maturity may differ substantially henrich2010. Second, pubertal timing was assessed via self-report (PDS), which may be less accurate than physical assessments or hormone assays. Third, we cannot rule out confounding by socioeconomic status or other third variables—though we did control for parent education and household income. Fourth, the longitudinal design, while superior to cross-sectional alternatives, cannot establish causation, and unmeasured confounds may account for the mediation finding. Fifth, the confidence intervals barely excluded zero for the mediation effect, suggesting the effect may not replicate.

Future research should consider several directions. First, neuroimaging studies examining whether pubertal timing predicts prefrontal cortex structure or function above and beyond chronological age would clarify the biological versus social pathways. Second, cross-cultural research examining whether the mediation effect holds in non-WEIRD samples would test generalizability. Third, longer follow-up into early adulthood would clarify whether timing effects emerge later—some studies suggest continued executive function development into the mid-20s. Fourth, alternative mediation models examining peer relationships, academic demands, or sleep patterns as pathways would extend the present findings. Fifth, making data and code openly available (as advocated by the open science movement) would enable replication and extension.

In conclusion, this longitudinal study found no evidence that pubertal timing directly influences executive function maturation during adolescence, with null effects across inhibitory control, working memory, and cognitive flexibility. However, perceived social maturity partially mediated the timing–executive function relationship, suggesting that psychosocial pathways may shape cognitive development even if biological pubertal timing does not. These findings qualify theoretical models emphasizing direct hormonal effects and highlight the potential role of social experience in executive function development.